Premankur Banerjee

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Ph.D. researcher specializing in Haptics, VR/AR, HCI and Robotics, with over 7 years of hands-on experience building end-to-end multisensory, immersive, and interactive systems. From hardware design and real-time control algorithms to software integration and extensive user studies, my work focuses on transforming prototypes into validated, scalable solutions, making bold ideas evolve into innovative prototypes and *impossible* interactions part of everyday technology.

EDUCATION

University of Southern California August 2023 – Present
Ph.D., Computer Science, GPA: 3.95 / 4.00 *Expected Graduation – **August 2027***

Technical University of Munich October 2019 – December 2021
M.Sc. (*with distinction*), Electrical and Computer Engineering, GPA: 1.4 / 1.0

University of Calcutta August 2015 – May 2019
B.Tech., Electronics and Communications Engineering, GPA: 9.16 / 10.00

EXPERIENCE

Graduate Research Assistant, Haptics Robotics and Virtual Interactions (HaRVI) Lab
Thomas Lord Department of Computer Science, University of Southern California, Los Angeles, CA August 2023 – Present

Surfing in Virtual Reality (Project Lead):

- Developed a VR surfing simulation integrating a 6-DoF motion platform in Unity 3D to study the effect of surf therapy on patients, promoting physical and psychological well-being.
- Implemented interaction techniques for navigating in VR via external tracking using depth cameras.
- Developed control optimization algorithms, real-time state estimation techniques, and optimized multi-DOF motion dynamics for improved force feedback, motion perception, and real-time physics control to increase simulation fidelity and user experience.
- Designed and engineered force-sensor platforms for bidirectional control, enabling maneuverability through center-of-pressure shifts and improving control accuracy and seamless user interactions in the simulator.
- Implemented multisensory feedback (airflow, vision, and motion) to increase immersion and reduce cybersickness, leading to a 35% improvement in user comfort and engagement.
- Designed and developed thermal and vibrotactile gloves to simulate water interaction while surfing in VR.
- Conducted controlled user studies with over 50 participants.

Multi-User Virtual System for Bidirectional, Mediated, Real-Time Social Touch (Project Lead):

- Developed a haptic-enabled multi-user VR and MR system for real-time, bidirectional social touch, enhancing presence and embodiment scores by 45%.
- Designed and built haptic gloves and sleeves with 52 vibrotactile (ERM) actuators in total.
- Implemented a WiFi-based cross-device communication protocol optimized for remote touch interactions with Arduino and Unity, reducing latency to under 100 ms.
- Conducted user studies demonstrating that visual-haptic feedback improved the pleasantness and realism of social interactions by 38% compared with visual-only interactions, leading to higher emotional engagement.

LLM-Driven Social Touch System for AI-Mediated Affective Haptic Interactions (Project Lead):

- Built a wireless AI-mediated social-touch system that maps natural language and affective state into real-time vibrotactile feedback through hand gloves and forearm sleeves with 52 ERMs in total.
- Developed a Python-Unity-Arduino runtime with a GUI for actuator testing, gesture authoring, simulation, real-hardware playback, and LLM-generated haptic control.
- Designed a closed-loop multimodal emotion-to-touch architecture combining VAD affect modeling, LLM-based gesture generation, safety clamping, social gating, and personalized user calibration.
- Benchmarked across 840 prompt-to-social touch generations, achieving 100% hardware validity and validating expressive control of haptic intensity and timing.

Redirected Walking (RDW) in Virtual Reality Using Ankle-Based Haptics (Project Co-Lead):

- Designed a wireless, gait-synchronized ankle-based vibrotactile system for RDW in VR, using four ERM actuators and two ankle-mounted IMUs to deliver swing-phase haptic cues.
- Conducted a user study with 28 participants across 14 RDW conditions, analyzing how vibration side, curvature direction, and gait-velocity-modulated feedback affect curvature-detection thresholds, presence, and simulator sickness.
- Demonstrated that unilateral ankle stimulation meaningfully improved RDW performance, increasing curvature-gain thresholds by up to 73.1% and enabling imperceptible redirection along tighter walking arcs, while movement-contingent feedback improved perceived naturalness, presence, and comfort.
- Derived design guidelines for scalable haptic RDW systems, showing that locomotor-state adaptation, spatial asymmetry, and gait-synchronized feedback are key to extending VR walking using lightweight wearable haptics.

Disability-Centered Vibrotactile System (Project Co-Lead, in collaboration with USC School of Cinematic Arts – Media Arts + Practice, Leonardo Crip Tech Incubator, and Narrative and Emerging Media Program, ASU):

- Built a wireless, modular vibrotactile wearable system with a real-time Unity control and authoring interface (GUI) for remote, multi-actuator haptic composition, recording, and playback.
- Led co-creative, user-centered system design and evaluation with Disabled artists, validating usability, engagement, and affective expressivity through controlled studies and live-performance deployment.

Research Engineer, Human-Computer Interface Group
Institut des Systèmes Intelligents et de Robotique (ISIR), Sorbonne Université, Paris, France June 2022 – May 2023

- Worked on real-time force-feedback control algorithms for a novel encounter-type haptic device delivering more than 100 N of kinesthetic feedback, enabling full-body interactions in room-scale VR environments.
- Designed and implemented dynamic VR scenarios integrating body-scale haptic interactions (e.g., pushing, leaning, and pulling) with motion tracking (OptiTrack) to enhance physical realism.
- Investigated low-cost haptic solutions by leveraging redirected walking and visio-haptic illusions, reducing the need for large-scale actuators while maintaining perceptual realism.
- Led experimental validation by conducting user studies.

Master's Thesis Student
German Aerospace Center (DLR), Munich, Germany May 2021 – November 2021

Multimodal VR Evaluation Suite for Novel Tactile Wearable Devices:

- Built a tactile-device-agnostic virtual environment for evaluating and comparing multiple tactile devices.
- Designed multiple interaction techniques for tactile modalities such as friction, temperature, texture, and shape.
- Conducted user studies with two novel tactile devices to validate the proposed evaluation suite.

Research Intern, Chair of Media Technology (LMT)
Technical University of Munich, Munich, Germany November 2019 – April 2021

- Developed an autoencoder-based compression technique to transmit vibrotactile signals over the internet.
- Simulated the Kinova MOVO robot in Unity 3D by connecting ROS and Unity for liquid-pouring tasks.
- Migrated Kinova MOVO to newer versions of Gazebo and ROS and developed autonomous grasping algorithms.

PUBLICATIONS

Journal Articles

- **P. Banerjee**, N. Li, J. Wang, H. Culbertson, Shaping Redirected Walking with Ankle-Based Vibrotactile Feedback, **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, presented at IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2026.
- **P. Banerjee**, S. Sweidan, C. Yunis, V. H. Cruz, H. Culbertson, *Crip Haptic Jam*: Vibrotactile Composition for Disability-Centered Dance, **IEEE Transactions on Haptics (ToH), Special Issue: Haptics for Arts**.
- **P. Banerjee**, M. Montiel, L. Tomita, O. Means, J. Kutch, H. Culbertson, The Impact of Airflow and Multisensory Feedback on Immersion and Cybersickness in a VR Surfing Simulation, **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, presented at IEEE VR 2025.

Conference Proceedings and Demos

- **P. Banerjee**, J. Wang, O. Means, H. Culbertson, SWIMVR: Simulating Water Interaction Using Multimodal Vibro-Thermal Rendering, **IEEE Haptics Symposium, 2026**.
- **P. Banerjee**, J. Wang, L. Tomita, M. Montiel, H. Culbertson, Virtual Encounters of the Haptic Kind: Towards a Multi-User VR System for Real-Time Social Touch, **2025 IEEE World Haptics Conference (WHC)**.
- **P. Banerjee**, J. Cherin, J. Upadhyay, J. Finley, J. Kutch, H. Culbertson, Perception and Control of Surfing in Virtual Reality Using a 6-DoF Motion Platform, in **Proceedings of EuroHaptics 2024**.
- H. Singh, L. P. Marcilla, **P. Banerjee**, M. Rothammer, T. Hulin, FerroVibe: Towards a Modular Tactile Device, **2024 IEEE Conference on Telepresence**.
- **P. Banerjee**, E. Muschter, H. Singh, B. Weber, T. Hulin, Towards a VR Evaluation Suite for Tactile Displays in Telerobotic Space Missions, **2023 IEEE Aerospace Conference**.
- H. Singh, T. Hulin, **P. Banerjee**, Rendering and Displaying Tactile Properties of Virtual Objects via ViESTac and FerroVibe, **2023 IEEE World Haptics Conference, Demos**.

Theses

- Banerjee, Premankur (2021), ViESTac: A Multimodal VR Evaluation Suite for Novel Tactile Devices. DLR-Interner Bericht, DLR-IB-RM-OP-2021-214. Master's thesis, Technical University of Munich, 104 pages.

PATENTS

- Heather Marie Culbertson, Premankur Banerjee, Jason James Kutch, Evelyne Falann Bernardette Morisseau. **A Virtual Reality Motion Platform**. US Application No. 63/868,024 | USC Ref. 2025-099-01; MCC Ref. 11760-012PV1, filed: August 21, 2025.

TEACHING EXPERIENCE

- Teaching Assistant, **CSCI-526: Advanced Mobile Devices and Game Consoles**, Fall 2026, USC.
- Teaching Assistant, **DSCI-517: Research Methods and Analysis for User Studies**, Fall 2024, USC.
- Teaching Assistant, **EI5052: Time-Varying Systems and Computation**, Winter 2021, TUM.
- Teaching Assistant, **EI72071: Computational Haptics Laboratory**, Summer 2021, TUM.
- Teaching Assistant, **EI7341: Image and Video Compression**, Spring 2021, TUM.
- Teaching Assistant, **EI70220: Digital Signal Processing**, Summer 2020, TUM.

HONORS, DISTINCTIONS, AND FELLOWSHIPS

- **Fellowships**
  - Link Foundation Fellowship in Modeling, Simulation, and Training, $35,000 (2026–2027).
  - Viterbi School of Engineering Fellowship, University of Southern California, $42,000 (2023–2024).
  - Deutschlandstipendium (German National Scholarship), Technical University of Munich, for academic performance and contributions to the scientific community, €7,200 (2020–2021).
- **Awards**
  - Undergraduate Mentor Award, Thomas Lord Department of Computer Science, Viterbi School of Engineering, University of Southern California (2025).
- **Press and Media Coverage**
  - Multi-user mediated social-touch project featured by USC News, TikTok/YouTube, and the official Arduino blog.
  - Research on Surfing in Virtual Reality featured by KTLA and published in USC Viterbi Magazine.
  - Disability focused work *Crip Haptic Jam* featured by Leonardo Crip Tech Incubator and Ability Central.
- **Science and Technology Outreach**
  - Demonstrated CoVR at Fête de la Science, contributing to public engagement in science and technology.

SKILLS

Technical Skills

- **Programming & Software:** C++, C, C#, Python, MATLAB, Unity 3D, Unreal Engine, CHAI3D, OpenHaptics, Arduino & Microcontroller programming, Blender, FreeCAD, ROS.
- **Haptics, VR, and Sensing:** Actuators, Force sensing and feedback, Tactile sensing, Wearables, Filtering, Temporal smoothing, Multiplayer VR/AR, Pressure Sensing, Affective haptics, Gesture-based Input, Motion tracking (OptiTrack, Intel RealSense, MediaPipe, BlazePose), 6-DoF Motion Platforms, Telepresence.
- **Simulation & Robotics:** MuJoCo, Gazebo, Physics modeling, OpenGL, Room-scale VR, Autonomous Robotics, Inverse kinematics, IMUs, Sensor Fusion, Wireless Communication Protocols, Teleoperation.
- **Machine Learning:** CNNs, Autoencoders, Signal Processing, Interactive Machine Learning, Large Language Models (LLM), Prompt Engineering, Multimodal Affect Recognition, PyTorch, TensorFlow, Transformers.
- **User Studies & Experimentation:** Experimental design, User-centered Design, Protocol creation, Data processing, and Statistical analyses (ANOVA, non-parametric tests, mixed models, and others); conducted over 200 user studies.

Soft Skills

- **Leadership & Teaching:** Mentored 20 graduate (M.S.) and undergraduate students at USC.
- **Collaboration & Communication:** Worked across interdisciplinary teams at USC, TUM, and Sorbonne University.
- **Project Management:** Led multiple projects and managed hardware-software integration in research environments.

ADDITIONAL INFORMATION

- **Languages:** English (native), Bengali (native), Hindi (native), German (professional), French (elementary).
- **Peer Reviewer:** IEEE Transactions on Haptics (ToH), IEEE Robotics and Automation Letters (RA-L), Springer Nature – Virtual Reality, IEEE World Haptics, IEEE Haptics Symposium, EuroHaptics, IEEE VR, IEEE ISMAR, Perspectives of Communication & Media (PCM), and Wisdom Academic Press.
- **Hobbies:** Acting, filmmaking, VFX design, reading, photography, singing, and film scoring.